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Institute of Technology and SportsComplex

Dept. of Electronics & Communication Engineering
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**Presentation on the topic : Design and Implementation of a Real Time Web Based Chat
Application Using MERN Stack and Socket.IO**

About :

Slide No.

02

Recent Advancements in the Field of Engineering and Technology (RAFET-2026)

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R_114

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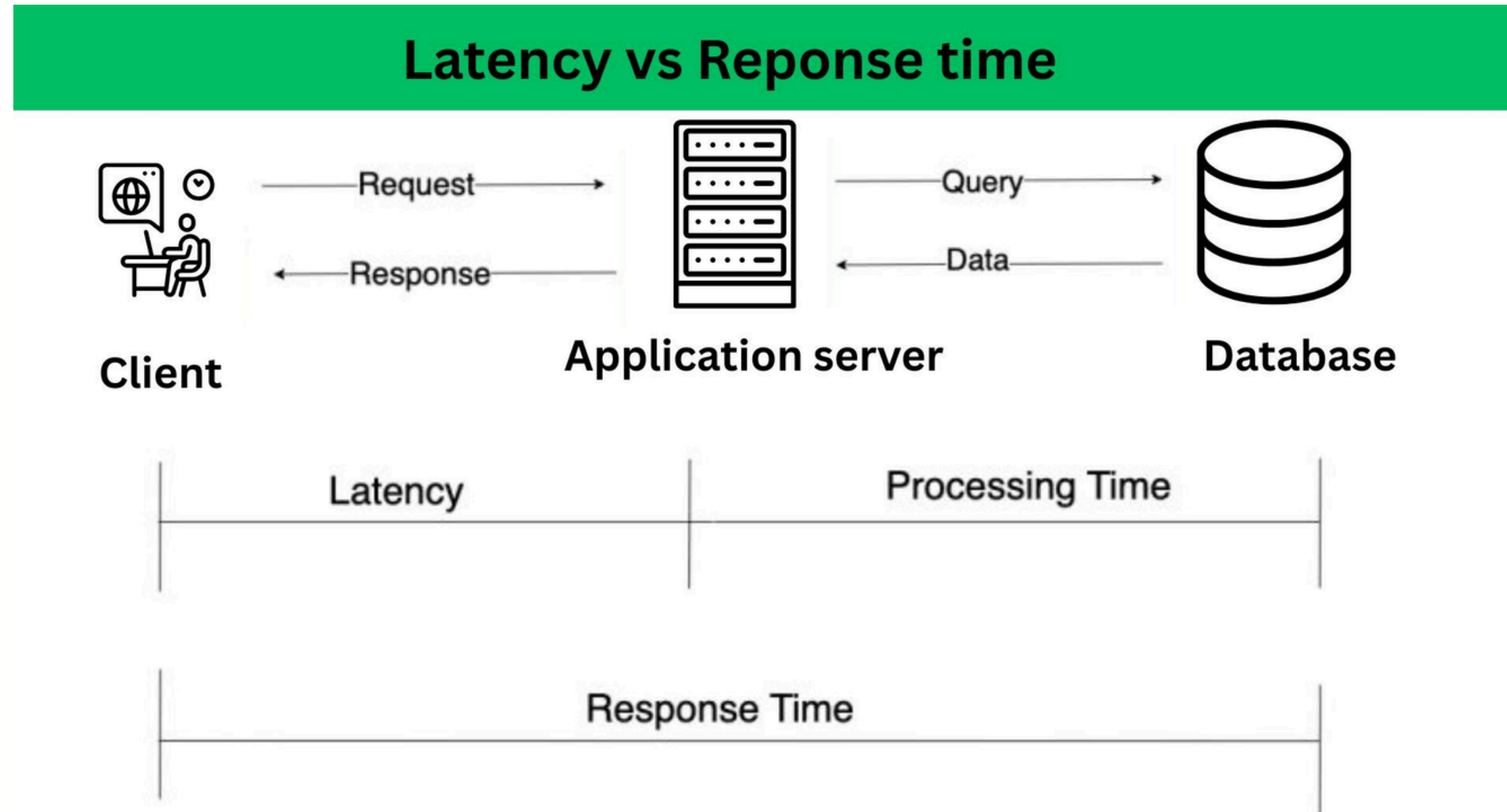
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Real-Time Chat Application using MERN Stack

Problem :

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05



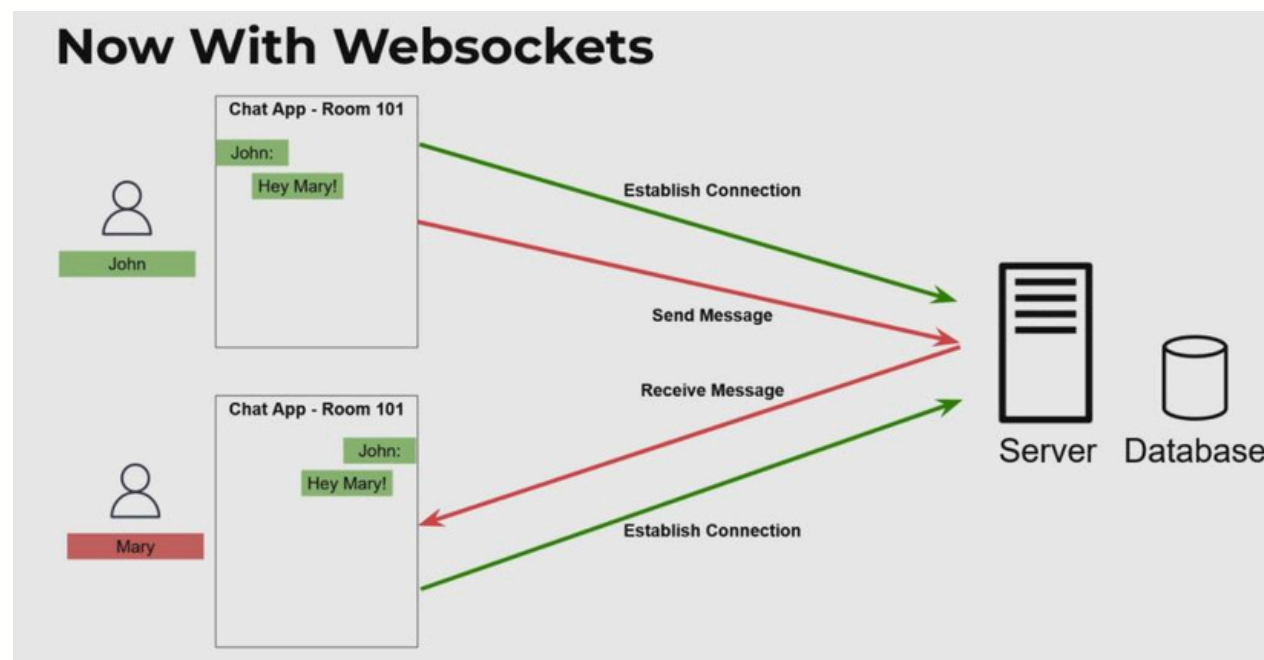
- **Delayed Communication**
- **No Real-Time**

Traditional apps depend on request-response, causing delays

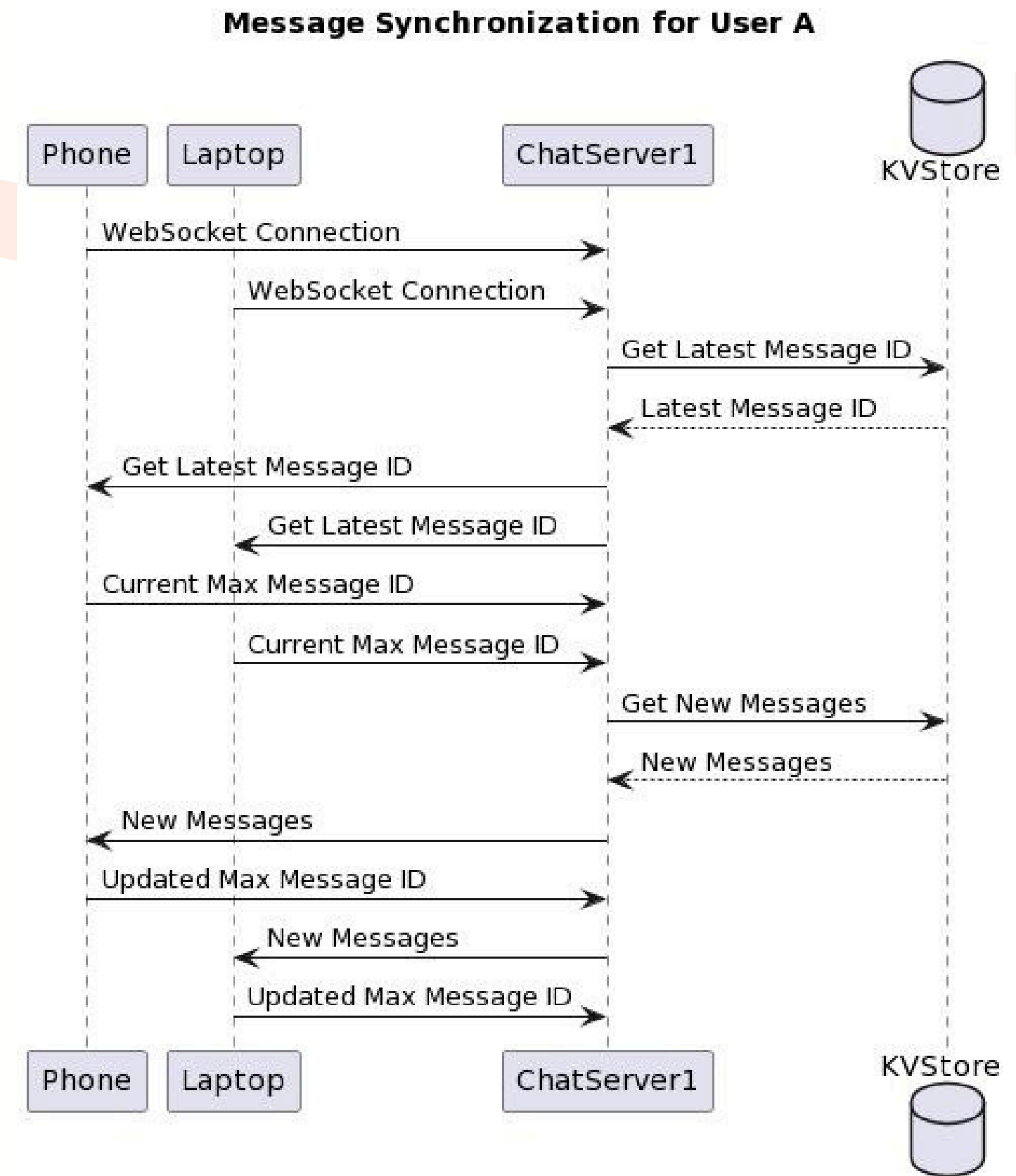
Solution :

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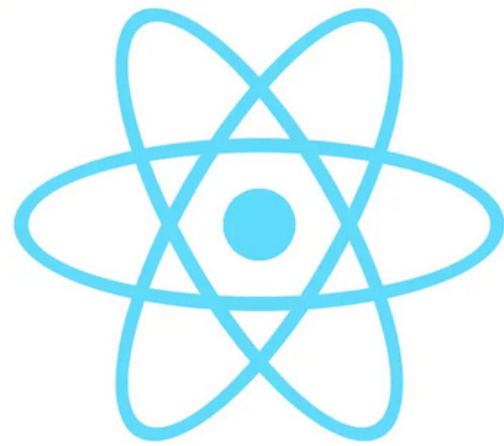
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- Real-Time Messaging
- Instant Updates



Tech Stack :



React JS

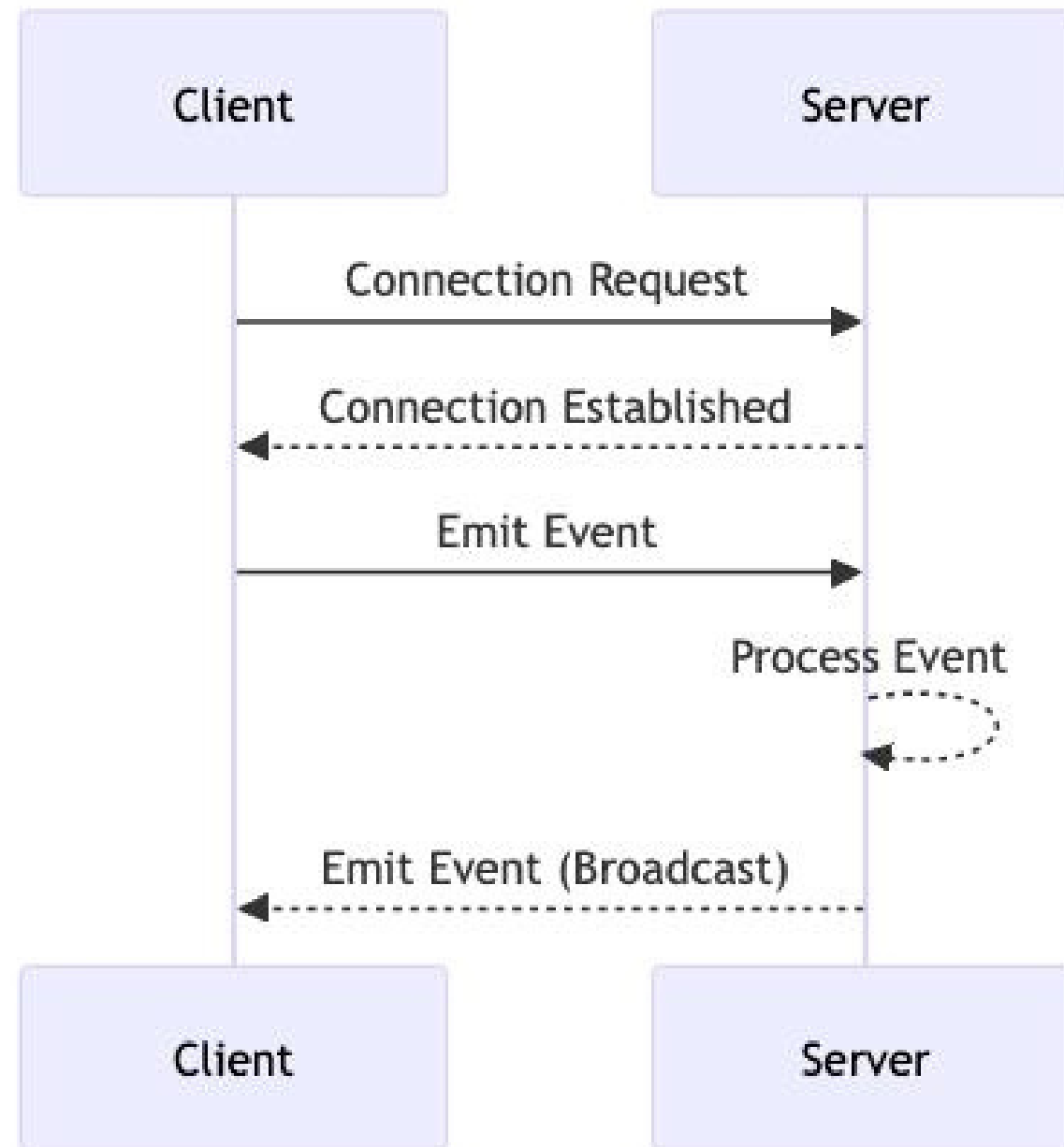
express.js



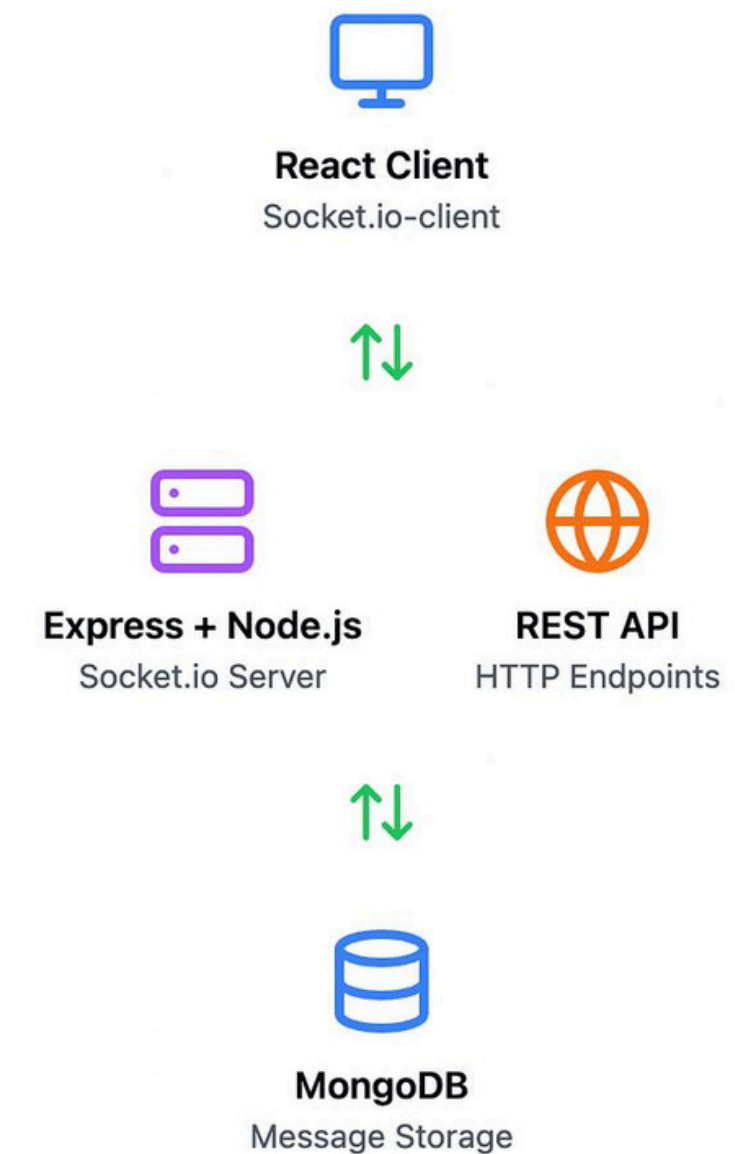
socket.io



Architecture :

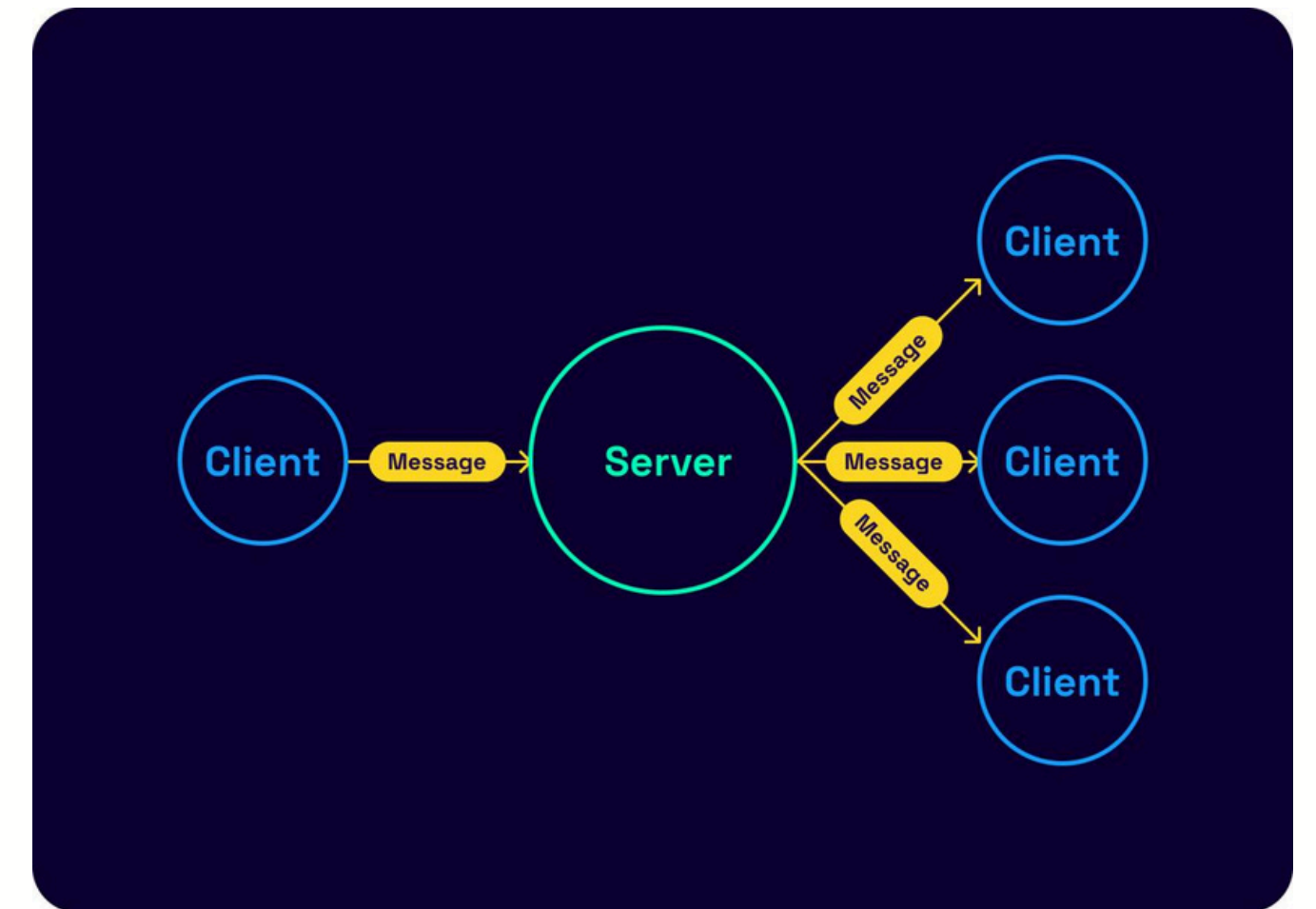
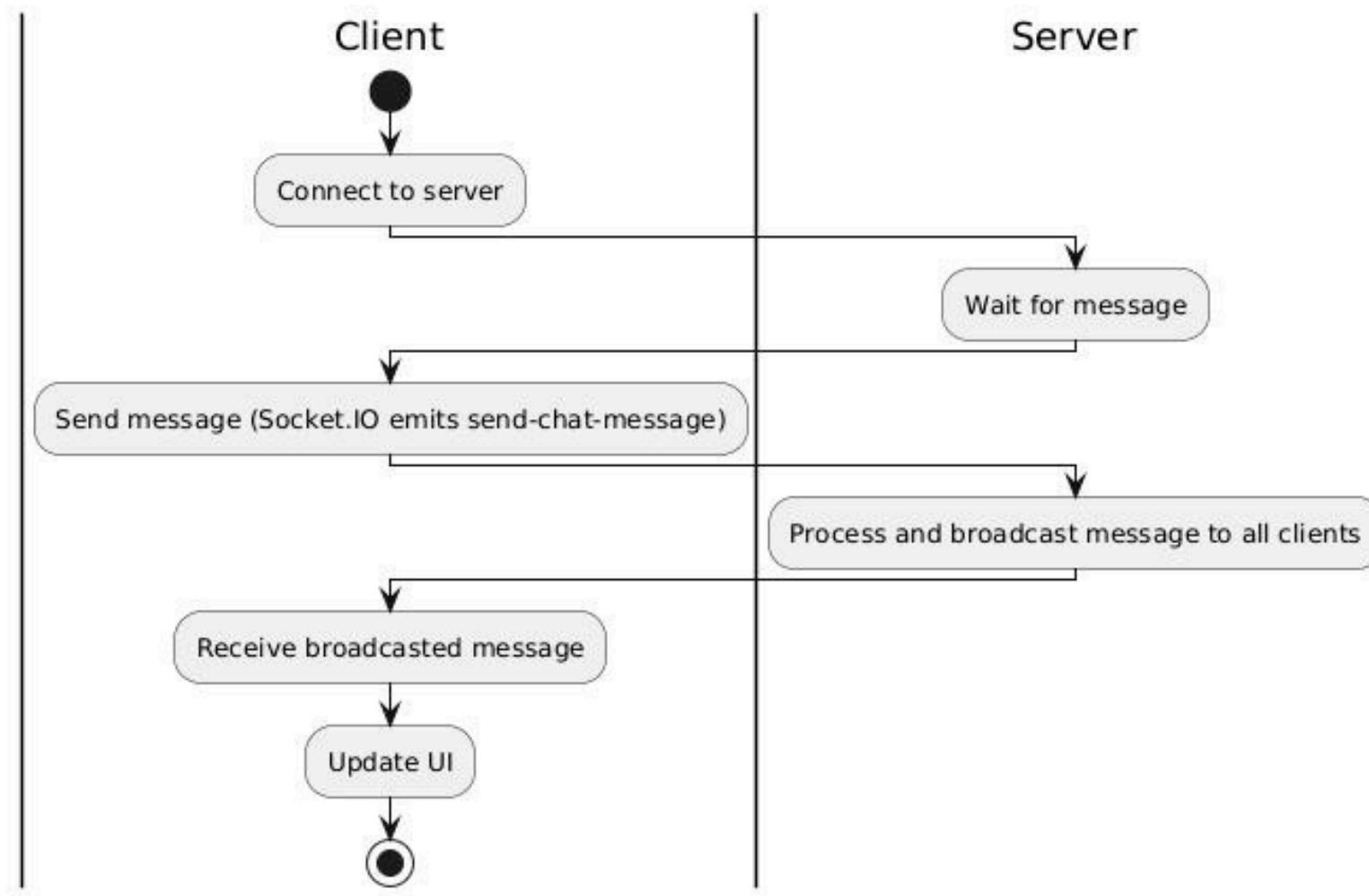


MERN + Socket.io Architecture



Client ↔ Server ↔ Database

Message Flow :



Demo / UI

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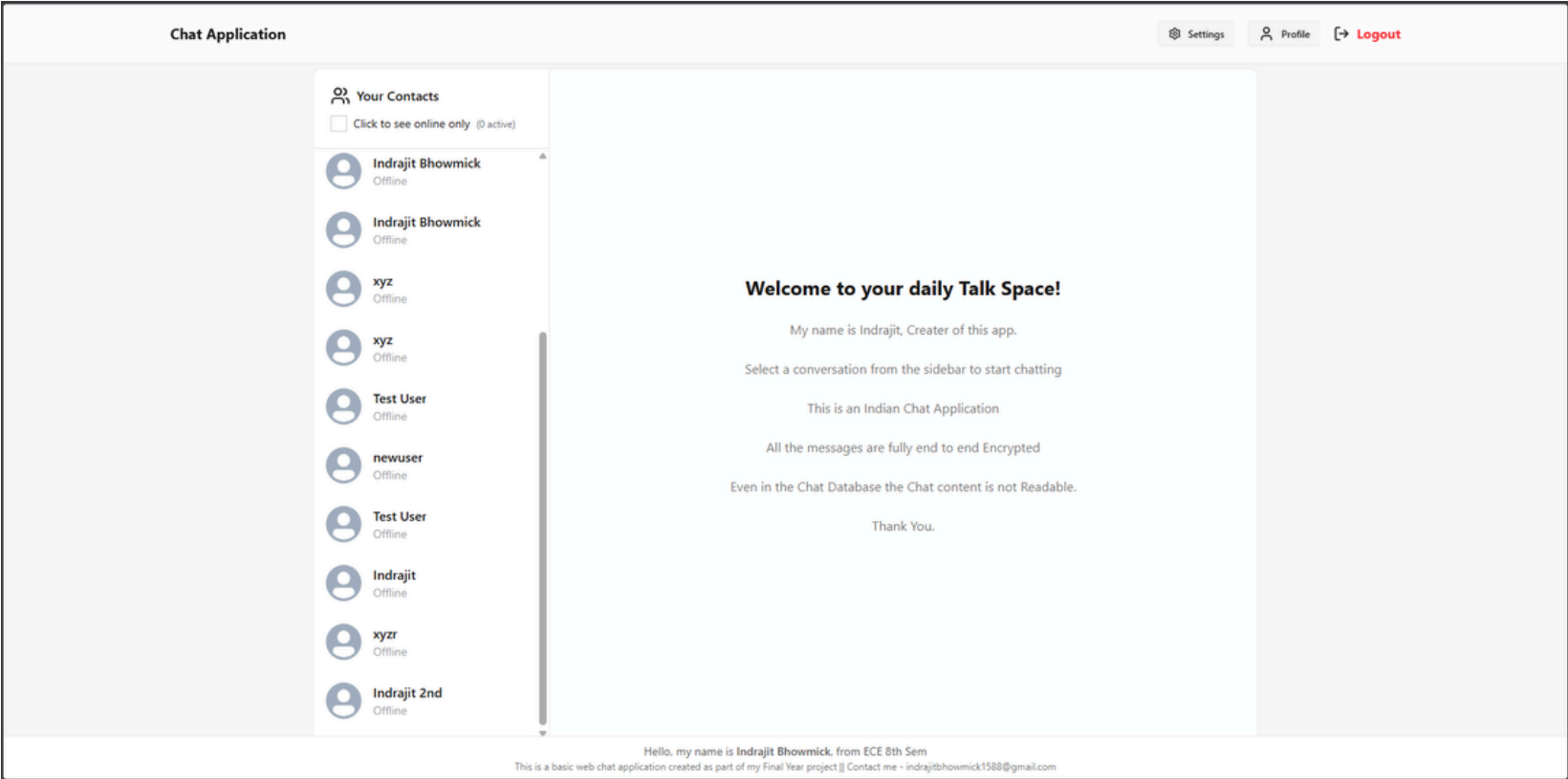
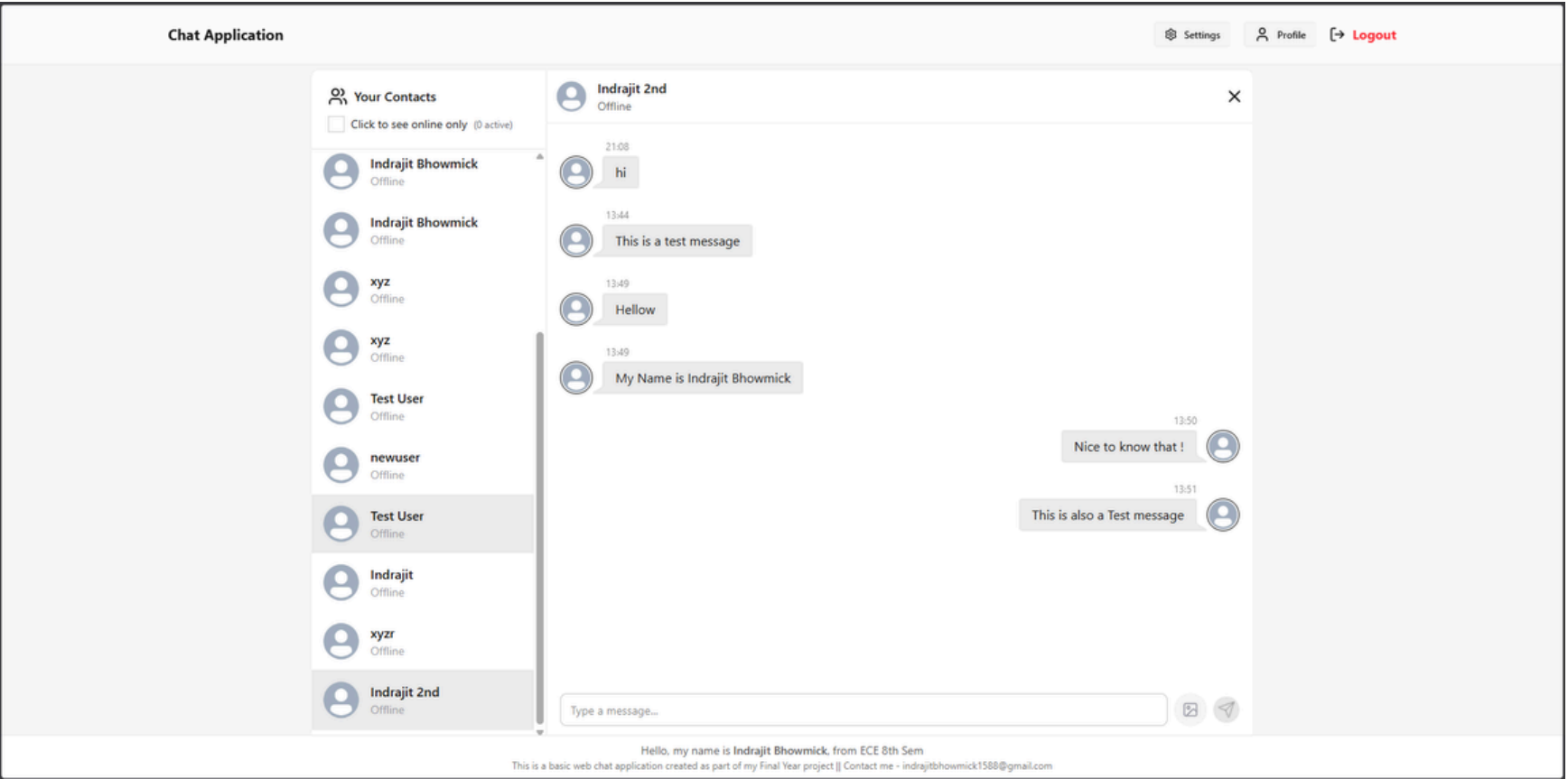


Figure 01 : Chat App Home Screen

Figure 02 : Chat App Messaging Screen



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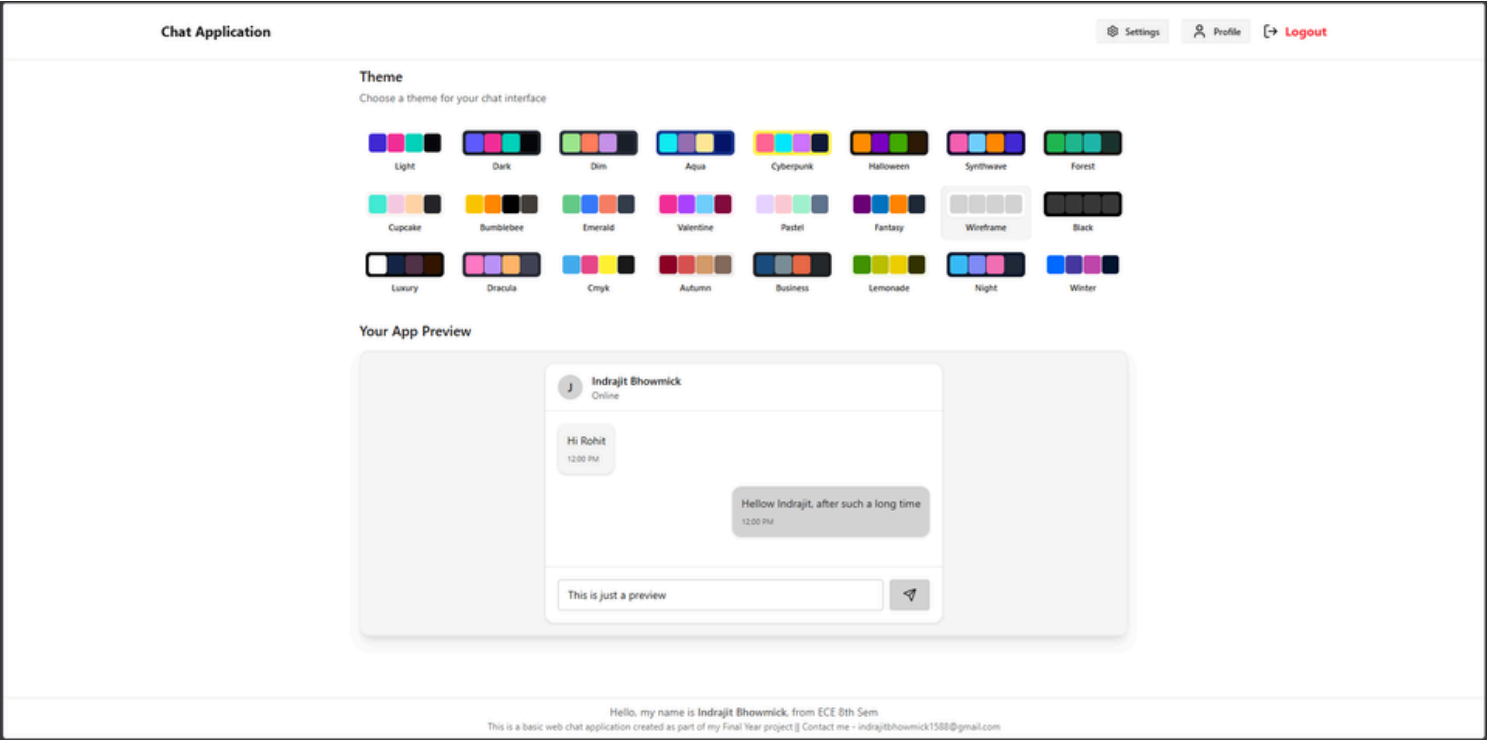


Figure 03 : Chat App Setting Page

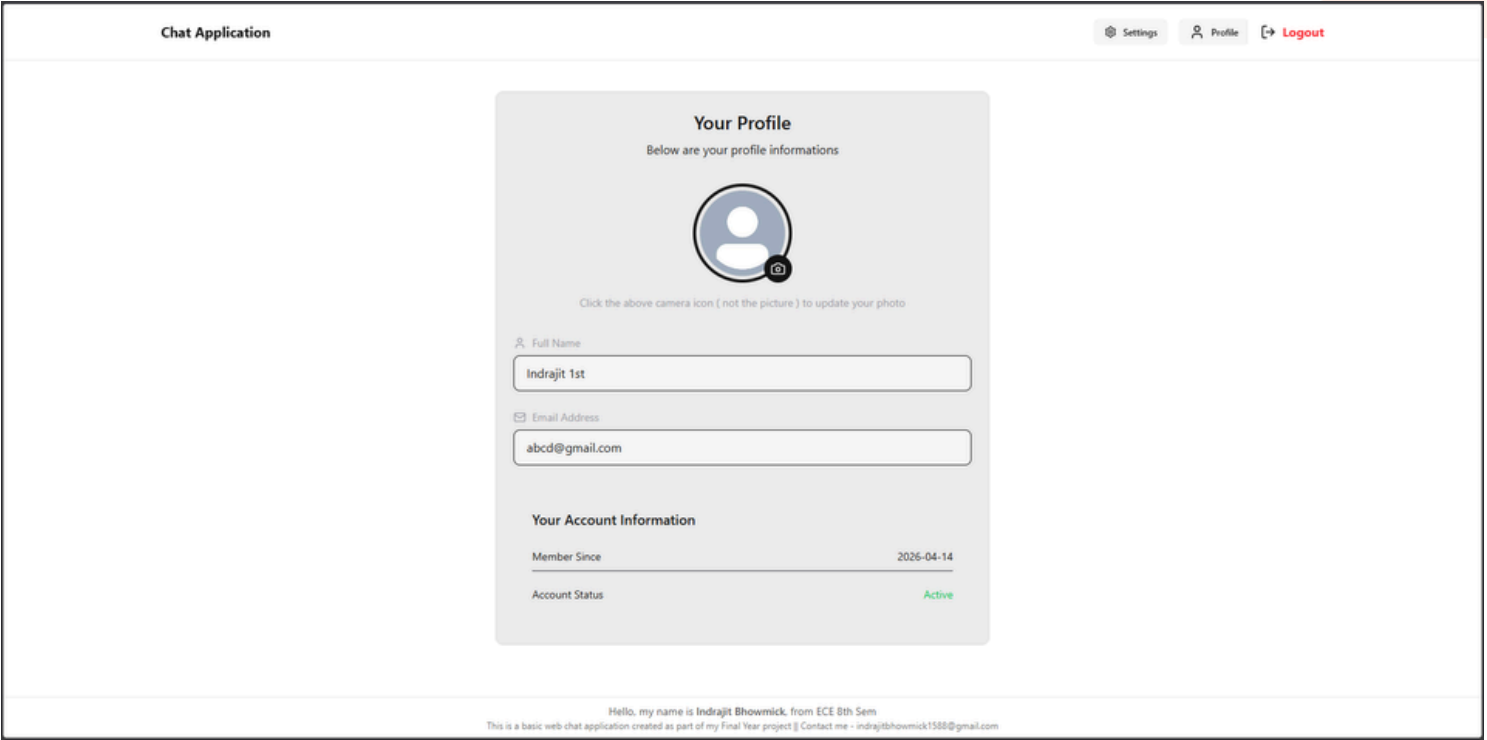


Figure 04 : Chat App Profile Page

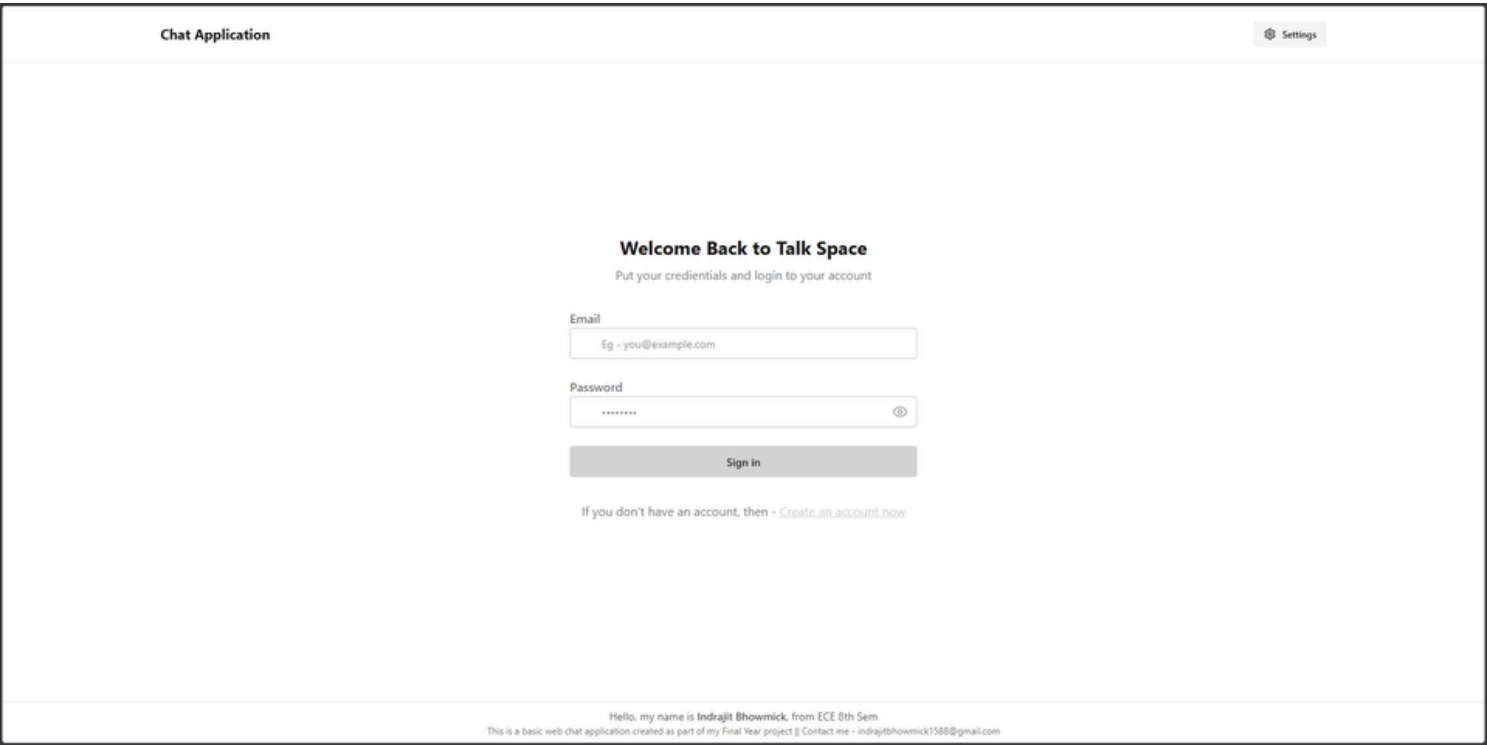


Figure 05 : Chat App Login Page

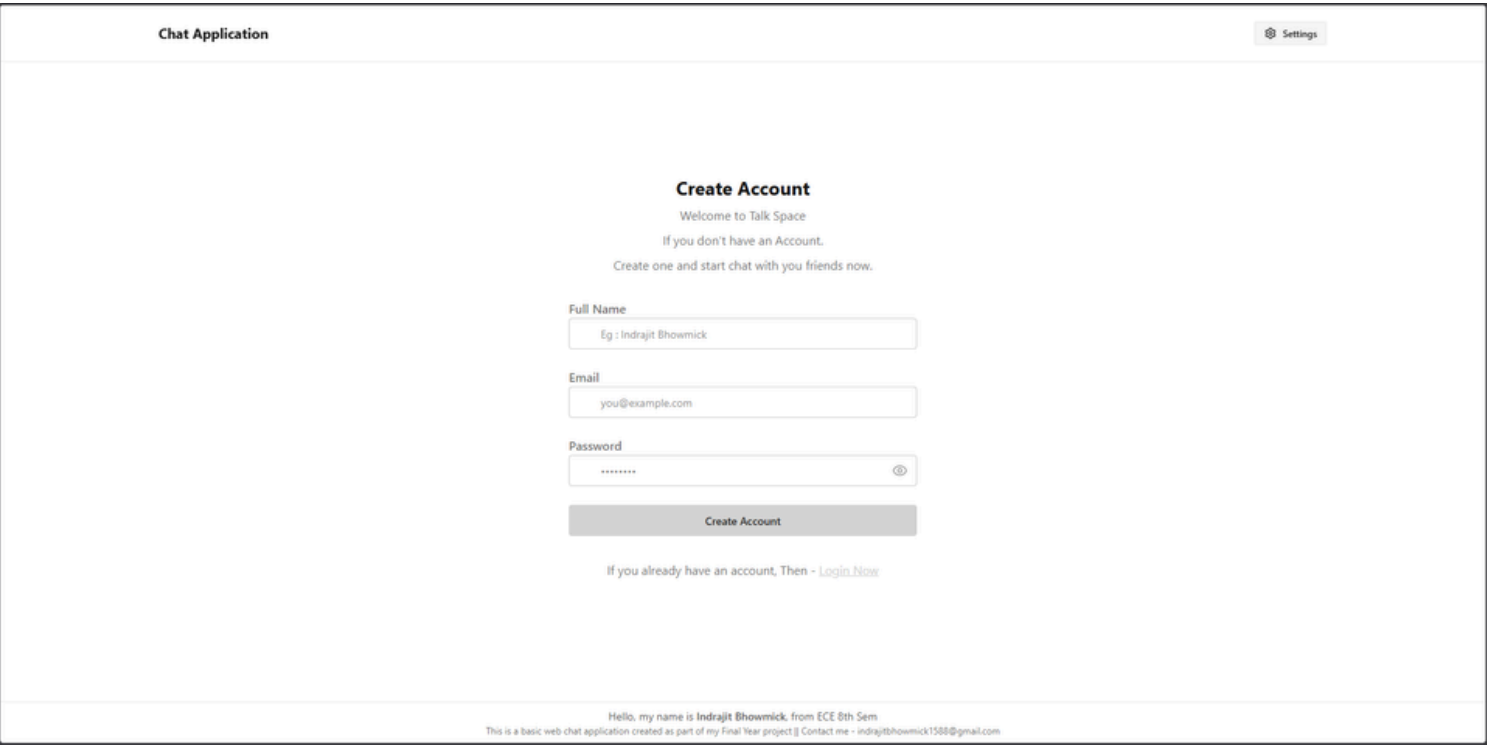


Figure 06 : Chat App Signup Page

Features :

- *Real-Time*
- *Multi-User*
- *Persistent Storage*
- *Responsive*

A Unique Idea :

- *Event-Driven System*
- *Persistent Connection*

Challenges :

- *Socket Issues*
- *Sync Problems*
- *Multi-User Handling*



Future Scope :

- *AI-based Smart Features*
- *Secure Payments Integration*
- *Media & File Sharing*
- *Video Calls*
- *Cloud Scaling*

Conclusion :

- *Real-Time Communication*
- *Event-Driven Architecture*
- *Scalable System*
- *Modern Web Stack*

Reference :

- I. Fette and A. Melnikov, “The WebSocket Protocol,” RFC 6455, 2011.
- “React Documentation,” Available: <https://react.dev>
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- “Socket.IO Documentation,” Available: <https://socket.io>
- MDN Web Docs, “WebSocket API,” Available: <https://developer.mozilla.org>

THANK YOU!

Feel free to approach if you have any questions.